

Alternative Data-Enhanced Credit Risk Scoring Using UPI Behavioral Analytics and Machine Learning

1. Abstract

Traditional credit-scoring systems primarily rely on financial history, repayment records, borrower demographics, income-related attributes, and existing credit information. Although these variables are effective for conventional credit-risk assessment, they may not fully capture a customer's ongoing digital financial behaviour.

This project presents a proof-of-concept credit-risk assessment framework combining conventional borrower information with alternative behavioral indicators derived from Unified Payments Interface (UPI) transaction data. A UPI transaction dataset containing 250,000 transactions was processed and transformed into behavioral indicators such as transaction frequency, average transaction amount, transaction volatility, weekend activity, merchant diversity, device diversity, and fraud exposure.

The machine-learning component was implemented in two stages. A Random Forest Classifier was used as the baseline model, achieving a ROC-AUC of **0.7183**, while the enhanced XGBoost framework achieved a ROC-AUC of **0.7457**, representing an absolute improvement of **0.0274** and a relative improvement of approximately **3.81%**. On the reported 160-sample classification evaluation, the enhanced model achieved **96.88% accuracy, 94.03% precision, 98.44% recall, and a 96.18% F1-score** for the Poor/risk class.

An interactive three-page Power BI dashboard was developed to visualize model performance, identify important credit-risk drivers, and analyze UPI behavioral characteristics. The project demonstrates a practical experimental framework for investigating the role of alternative digital-payment information in credit-risk analytics.

Keywords: Credit Risk, Machine Learning, UPI, Alternative Data, Behavioral Analytics, Random Forest, XGBoost, ROC-AUC, Feature Engineering, Power BI

2. Introduction

Credit-risk assessment is an essential function within banking and financial services. Financial institutions evaluate the creditworthiness of borrowers to estimate the probability that they may fail to meet their repayment obligations.

Traditional credit-scoring systems generally rely on variables such as repayment history, existing debt, income, employment information, loan characteristics, demographic information, and external credit indicators. These variables provide important information about a borrower's financial position and historical credit behaviour.

However, the rapid adoption of digital payment systems has created new sources of financial behavioral information. In India, UPI has become a major platform for everyday digital transactions. Transaction patterns generated through such systems can provide information about spending behaviour, transaction frequency, transaction stability, merchant interaction, fraud exposure, and payment timing.

Such behavioral information can potentially complement traditional credit variables, particularly in applications involving digital lending, behavioral risk profiling, and customers with limited conventional credit histories.

This project investigates a proof-of-concept framework in which behavioral features derived from UPI transactions are incorporated into a machine-learning-based credit-risk analysis pipeline. The project combines data analytics, behavioral feature engineering, machine learning, model evaluation, feature-importance analysis, and business intelligence visualization.

3. Problem Statement

Traditional credit-risk models primarily depend on conventional financial and borrower information and may not capture a customer's ongoing digital transaction behaviour.

Digital-payment data contains additional behavioral information that could potentially supplement conventional credit-risk indicators.

The objective of this project is therefore to develop a **proof-of-concept alternative-data credit-risk framework** that:

1. Analyzes conventional borrower and UPI transaction datasets.
2. Processes raw digital-payment transaction information.
3. Engineers meaningful behavioral variables from UPI activity.
4. Develops a baseline credit-risk classification model.
5. Develops an enhanced machine-learning framework incorporating behavioral information.
6. Compares the predictive performance of the baseline and enhanced frameworks.
7. Identifies the variables contributing to credit-risk predictions.
8. Analyzes important UPI behavioral characteristics.
9. Presents the results through an interactive Power BI dashboard.

4. Project Architecture

The project was implemented as a multi-stage data analytics and machine-learning pipeline.

The overall workflow is:

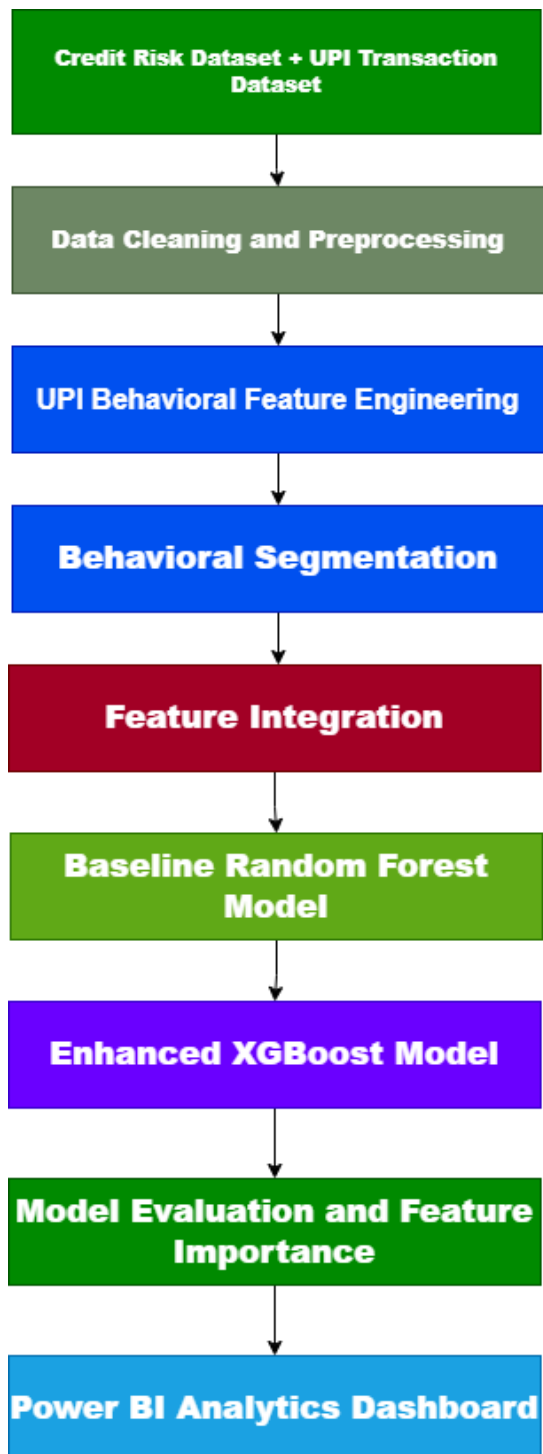


Figure 1: Overall Architecture of the Alternative Data Credit-Risk Framework

The architecture separates transaction-level behavioral analytics from the machine-learning modelling stage while allowing the derived behavioral variables to contribute to the enhanced credit-risk framework.

5. Dataset Description

5.1 Credit Risk Dataset

The primary credit dataset contains borrower-level financial, demographic, employment, and credit-related information suitable for supervised credit-risk modelling.

The original application dataset contained approximately:

307,511 borrower records

and approximately:

130 attributes

The dataset includes variables relating to borrower demographics, employment characteristics, loan information, financial information, external credit indicators, and other credit-risk attributes.

The target variable represents the credit/default-risk outcome required for supervised classification.

Following preprocessing and preparation of the modelling dataset, approximately **245K borrower records** were available in the final modelling data used in the Power BI analysis.

5.2 UPI Transaction Dataset

The alternative-data component of the project used a UPI transaction dataset containing:

250,000 transaction records

and:

17 original attributes

The attributes included:

- Transaction ID
- Timestamp
- Transaction type
- Merchant category
- Transaction amount in INR
- Transaction status
- Sender age group
- Receiver age group
- Sender state

- Sender bank
- Receiver bank
- Device type
- Network type
- Fraud flag
- Hour of day
- Day of week
- Weekend indicator

Unlike the credit dataset, this dataset initially represented individual transactions rather than borrower-level information.

Therefore, transaction-level information was processed and aggregated to derive higher-level behavioral indicators suitable for analytics and modelling.

6. Data Preprocessing and Analytics

Data preprocessing was performed before feature engineering and machine-learning implementation to ensure that the datasets were suitable for subsequent analysis.

Python and Pandas were primarily used for transaction-level processing.

The major preprocessing activities included:

- Dataset inspection
- Verification of dataset dimensions
- Column-name standardization
- Data-type verification
- Missing-value inspection
- Duplicate checking
- Transaction-level validation
- Identification of relevant behavioral variables
- Aggregation of transaction information
- Preparation of model-compatible features

The UPI dataset contained **250,000 rows and 17 original columns** after loading.

Following preprocessing, the dataset retained the required transaction information for behavioral feature engineering.

7. UPI Behavioral Feature Engineering

One of the major data-analytics components of the project involved transforming raw transaction information into interpretable behavioral indicators.

Individual UPI transactions were aggregated to create approximately:

400 behavioral segments

These segments summarized different patterns of digital-payment behaviour.

The major engineered features were:

Behavioral Feature	Description
Total Transactions	Number of transactions associated with a behavioral segment
Average Transaction Amount	Average monetary value of transactions
Transaction Volatility	Variation in transaction amounts within a segment
Fraud Ratio	Proportion of transactions associated with the fraud indicator
Weekend Ratio	Proportion of transactions performed during weekends
Merchant Diversity	Number/diversity of merchant interactions
Device Diversity	Diversity of devices associated with transactions

These engineered variables transformed transaction-level information into features capable of representing broader digital financial behaviour.

For example, **transaction volatility** provides information about the stability or variability of transaction amounts, while **total transactions** provides an indication of transaction activity.

Similarly, weekend ratio captures temporal behaviour, while fraud ratio provides information regarding exposure to transactions identified by the dataset's fraud indicator.

8. Machine Learning Methodology

The machine-learning component was implemented using two classification frameworks.

The first served as the baseline model, while the second represented the enhanced modelling framework.

8.1 Baseline Model — Random Forest Classifier

The baseline credit-risk model was developed using a **Random Forest Classifier**.

Random Forest is an ensemble machine-learning algorithm that constructs multiple decision trees and combines their predictions to generate the final classification output. By combining several decision trees, Random Forest can capture nonlinear relationships and interactions between variables while reducing dependence on an individual tree.

The baseline model primarily represents the conventional credit-risk modelling stage and provides a reference performance against which the enhanced framework can be compared.

Machine-Learning Algorithm: Random Forest Classifier

Baseline ROC-AUC: 0.7183

8.2 Enhanced Model — XGBoost Classifier

The enhanced model was developed using an **XGBoost Classifier (Extreme Gradient Boosting)**.

XGBoost is a gradient-boosting machine-learning algorithm that sequentially constructs decision trees. Each subsequent tree attempts to improve the errors made by the preceding trees, allowing the model to learn complex nonlinear relationships and interactions between variables.

The enhanced modelling framework incorporated the engineered UPI behavioral indicators alongside the conventional credit-risk information.

Behavioral variables included:

- Transaction Volatility
- Average Transaction Amount
- Fraud Ratio
- Total Transactions
- Weekend Ratio
- Merchant Diversity
- Device Diversity

Machine-Learning Algorithm: XGBoost Classifier

Enhanced ROC-AUC: 0.7457

9. Model Evaluation

The primary metric used for comparing model performance was Receiver Operating Characteristic – Area Under the Curve (ROC-AUC).

ROC-AUC measures the ability of a classification model to distinguish between the two target classes across different classification thresholds. A higher ROC-AUC generally indicates stronger discriminatory capability.

9.1 Model Performance Comparison

The observed results were:

Model	Algorithm	ROC-AUC
Baseline Model	Random Forest Classifier	0.7183
Enhanced Model	XGBoost Classifier	0.7457
Absolute ROC-AUC Gain	—	+0.0274
Relative ROC-AUC Improvement	—	+3.81%

The enhanced framework achieved an absolute ROC-AUC increase of 0.0274, equivalent to 2.74 percentage points, and a relative improvement of approximately 3.81% compared with the baseline model.

9.2 Classification Performance

In addition to ROC-AUC, the enhanced XGBoost model was evaluated using a confusion matrix containing 160 evaluation samples.

The confusion matrix produced the following results:

- 92 Good cases were correctly classified as Good.
- 63 Poor cases were correctly classified as Poor.
- 4 Good cases were incorrectly classified as Poor.
- Only 1 Poor case was incorrectly classified as Good.

Therefore, the model correctly classified 155 of the 160 evaluation samples.

The classification metrics derived from the confusion matrix are:

Metric	Result
Accuracy	96.88%
Precision (Poor Class)	94.03%
Recall / Sensitivity (Poor Class)	98.44%
F1-Score (Poor Class)	96.18%
Specificity	95.83%

The enhanced model achieved an overall classification accuracy of 96.88%. For the Poor/risk class, it achieved a precision of 94.03% and recall of 98.44%, resulting in an F1-score of 96.18%.

The high recall is particularly relevant in a credit-risk context because the model correctly identified 63 of the 64 actual Poor cases, with only one Poor case incorrectly classified as Good.

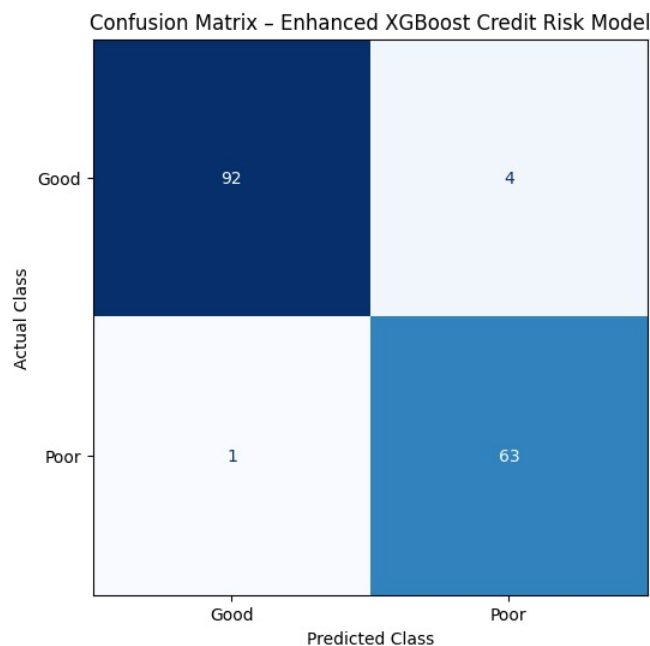


Figure 2: Confusion Matrix of the Enhanced XGBoost Credit Risk Model

9.3 Methodological Consideration

An important methodological consideration is that both the feature set and machine-learning algorithm changed between the baseline and enhanced frameworks.

The baseline model uses a Random Forest Classifier with the conventional credit-risk framework, whereas the enhanced model uses an XGBoost Classifier together with additional behavioral information.

Consequently, the improvement from 0.7183 to 0.7457 ROC-AUC should not be attributed exclusively to the addition of UPI behavioral features. Part of the performance difference may also result from the change in machine-learning algorithm.

Therefore, the results demonstrate that the enhanced framework combining behavioral information and XGBoost outperformed the Random Forest baseline within the experimental setup, rather than establishing that behavioral variables alone caused the observed performance improvement.

This version is cleaner for the final report because it gives the reader the overall model comparison first, then dives into the enhanced model's confusion matrix.

10. Feature Importance Analysis

Feature-importance analysis was performed to understand which variables contributed strongly to the predictions generated by the machine-learning framework.

The analysis showed that conventional credit-related variables remained among the strongest predictors.

In particular, external credit indicators such as:

EXT_SOURCE_2

and

EXT_SOURCE_3

were among the most influential overall features.

This suggests that established credit information continues to provide substantial predictive information for borrower-risk assessment.

However, behavioral features also contributed to the enhanced framework.

Important UPI behavioral variables included:

- Transaction Volatility
- Average Transaction Amount
- Fraud Ratio
- Total Transactions
- Weekend Ratio
- Merchant Diversity

Among the behavioral variables, **transaction volatility** showed relatively high feature importance.

These results support the concept of using alternative transaction information as a **supplement to conventional credit information**, rather than necessarily replacing established credit-risk variables.

11. Power BI Dashboard

An interactive Power BI dashboard was developed to present the results of the data analytics and machine-learning components in an interpretable format.

The dashboard contains three pages:

1. Executive Summary
2. Credit Risk Drivers and Behavioral Feature Importance
3. UPI Behavioral Analytics

11.1 Executive Summary Dashboard

The first dashboard page provides a high-level overview of the project and machine-learning performance.

The dashboard displays:

- Approximately 245K modelling borrowers
- 400 behavioral segments
- Baseline ROC-AUC
- Enhanced ROC-AUC
- ROC-AUC gain
- Baseline versus enhanced model-performance comparison

The page allows users to quickly understand the scale of the modelling dataset and the difference in performance between the two machine-learning frameworks.



Figure 3: Executive Summary – Alternative Data Credit Scoring Dashboard

11.2 Credit Risk Drivers and Behavioral Feature Importance

The second Power BI page focuses on model explainability and feature importance.

The first visualization displays the **Top 15 Credit Risk Drivers**, allowing the most influential variables in the model to be identified.

The second visualization focuses specifically on the engineered UPI behavioral features.

The behavioral variables displayed include transaction volatility, average transaction amount, fraud ratio, total transactions, weekend ratio, and merchant diversity.

This separation allows traditional credit-risk drivers and alternative behavioral indicators to be interpreted independently.

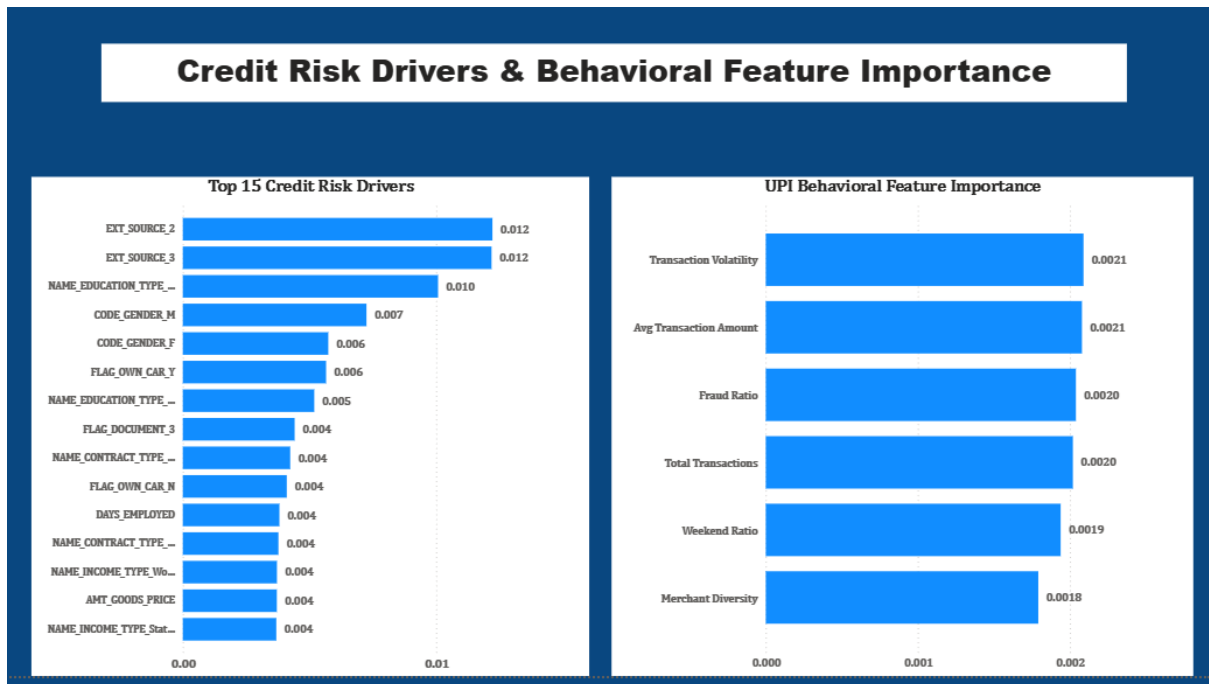


Figure 4: Credit Risk Drivers and UPI Behavioral Feature Importance

The analysis demonstrates that conventional credit indicators remain dominant while behavioral variables provide supplementary information within the enhanced modelling framework.

11.3 UPI Behavioral Analytics

The third Power BI page focuses specifically on behavioral analytics derived from UPI transaction information.

The primary behavioral KPIs include:

Average Transaction Amount: approximately ₹1.29K

Average Transaction Volatility: approximately 1.79K

Weekend Transaction Ratio: approximately 28.46%

Average Fraud Ratio: approximately 0.19%

Two distribution visualizations were also created.

Transaction Activity Distribution

This visualization shows the distribution of behavioral segments according to their total transaction activity.

It provides insight into how transaction frequency varies across the behavioral groups.

Transaction Volatility Distribution

This visualization represents the distribution of transaction volatility across behavioral segments.

It provides information about how strongly transaction amounts vary within different behavioral patterns.

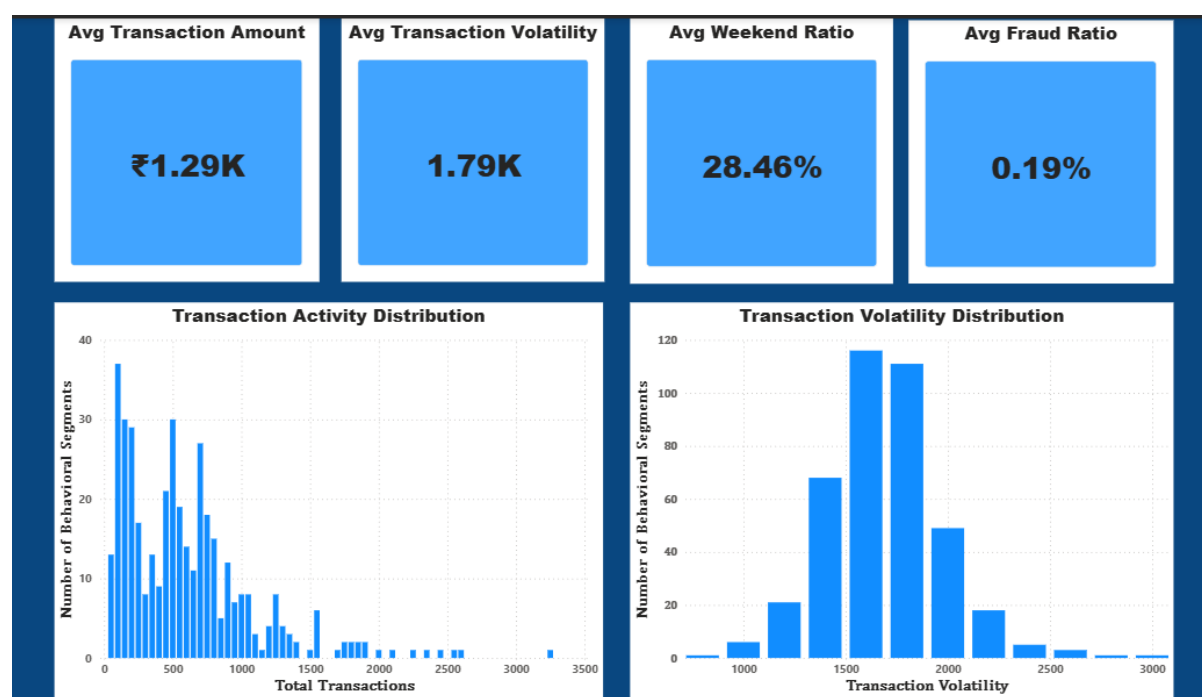


Figure 5: UPI Behavioral Analytics Dashboard

12. Key Findings

The baseline Random Forest model achieved a ROC-AUC of **0.7183**, while the enhanced XGBoost framework achieved a ROC-AUC of **0.7457**. This represents an absolute ROC-AUC gain of **0.0274**, equivalent to a relative improvement of approximately **3.81%** over the baseline.

On the reported **160-sample evaluation set**, the enhanced XGBoost model correctly classified **155 samples**, resulting in an overall accuracy of **96.88%**.

For the Poor/risk class, the enhanced model achieved **94.03% precision**, **98.44% recall**, and a **96.18% F1-score**. Of the 64 actual Poor cases, **63 were correctly identified**, while only one was incorrectly classified as Good. The model also achieved a specificity of **95.83%**.

Traditional credit variables, particularly external credit indicators such as **EXT_SOURCE_2** and **EXT_SOURCE_3**, remained among the strongest overall predictors.

Among the engineered UPI behavioral variables, **transaction volatility** demonstrated relatively high feature importance. Average transaction amount, fraud ratio, total transactions, weekend activity, and merchant diversity also provided supplementary behavioral information.

Behavioral analytics showed an average transaction amount of approximately **₹1.29K**, average transaction volatility of approximately **1.79K**, weekend transaction ratio of **28.46%**, and average fraud ratio of approximately **0.19%**.

Overall, the experimental results demonstrate that the enhanced XGBoost and behavioral-data framework provided stronger predictive performance than the Random Forest baseline.

13. Business Relevance

The project has potential relevance to banks, digital lenders, fintech companies, consulting firms, and other organizations involved in financial-risk analytics.

Traditional credit assessment can be challenging for individuals with limited historical credit information. Digital transaction behaviour may provide an additional source of information for understanding such customers.

Potential applications of the proposed framework include:

- Credit-risk assessment
- Digital lending
- Thin-file customer analysis
- Alternative credit scoring
- Behavioral risk profiling
- Early-risk identification
- Customer financial-behavior analysis
- Supplementary credit underwriting
- Fintech risk analytics

For example, transaction volatility and transaction frequency may provide information about the consistency of a customer's financial behaviour, while fraud-related indicators could provide supplementary information for risk monitoring.

The intended approach is not to replace conventional credit information but to investigate whether behavioral information can **complement established credit-risk assessment systems**.

14. Project Limitations

Although the proposed framework demonstrates the potential of combining conventional credit-risk information with UPI-derived behavioral analytics, several limitations must be considered while interpreting the results.

The primary limitation is that the **UPI transaction dataset and the credit-risk dataset do not represent the same real-world individuals**. The behavioral information was therefore incorporated as part of a proof-of-concept experimental framework rather than through direct customer-level linkage. Consequently, the results should not be interpreted as the performance of a production-ready banking credit-scoring system.

Another important methodological limitation is that the baseline and enhanced frameworks use different machine-learning algorithms as well as different feature configurations. The baseline model uses a **Random Forest Classifier**, whereas the enhanced framework uses an **XGBoost Classifier** together with behavioral information. Therefore, the observed increase in ROC-AUC from **0.7183 to 0.7457** cannot be attributed exclusively to the addition of UPI behavioral features. A portion of the improvement may also result from differences between the Random Forest and XGBoost algorithms.

Furthermore, the reported confusion-matrix metrics are based on an evaluation set containing **160 samples**. Although the enhanced model achieved strong classification performance on this evaluation set, including **96.88% accuracy and 98.44% recall**, these values should be interpreted within the scope and size of the reported evaluation.

Other limitations of the project include:

- Synthetic or indirect integration between the two datasets
- Lack of direct customer-level linkage
- Limited transaction-history duration
- Dataset-specific behavioral distributions
- Credit-risk class imbalance
- Low occurrence of fraud events
- Absence of real-time transaction streams
- Lack of production banking data
- Absence of production-level fairness and bias evaluation
- No regulatory or operational validation

A real-world implementation would require correctly linked and anonymized customer data, appropriate customer consent, privacy and security controls, fairness evaluation, extensive model validation, and compliance with applicable financial regulations.

15. Future Scope

The current project provides a foundation that can be extended in several directions to develop a more comprehensive credit-risk analytics system.

15.1 Real Customer-Level Data Integration

A major extension would involve using anonymized datasets in which conventional credit information and digital transaction behaviour correspond to the same customers. This would enable direct evaluation of whether UPI behavioral variables provide incremental predictive value beyond traditional credit information.

15.2 Controlled Model Comparison

Future experiments should use a controlled modelling framework consisting of:

- Random Forest without behavioral features
- Random Forest with behavioral features
- XGBoost without behavioral features
- XGBoost with behavioral features

Such an experiment would allow the effect of the machine-learning algorithm to be separated from the effect of adding alternative behavioral features.

15.3 Explainable Artificial Intelligence

Explainability techniques such as SHAP (SHapley Additive exPlanations) could be incorporated to understand how individual variables influence each borrower's predicted credit-risk score.

This would provide more detailed model interpretation than global feature-importance scores alone.

15.4 Temporal Behavioral Analysis

The current behavioral indicators are primarily aggregated measures. Future systems could analyze how customer transaction behaviour changes over time.

Temporal features could include changes in transaction frequency, sudden increases in spending volatility, variations in merchant behaviour, and changes in transaction timing.

15.5 Dynamic Credit-Risk Scoring

A dynamic credit-scoring system could periodically update a customer's risk estimate based on recent transaction behaviour rather than relying entirely on a static credit assessment.

15.6 Real-Time Risk Monitoring

Streaming transaction information could be incorporated into the framework to identify significant behavioral changes and update risk indicators in near real time.

15.7 Model Drift Detection

Future implementations could continuously monitor changes in data distributions and model performance to detect model drift and determine when model retraining is required.

15.8 Fairness and Bias Analysis

Credit-risk models should be evaluated for potential bias across relevant customer groups. Fairness metrics and responsible-AI techniques could therefore be incorporated before considering real-world deployment.

15.9 Cloud and API Deployment

The complete analytics and machine-learning pipeline could be deployed on cloud infrastructure for scalable data processing and model inference.

The trained model could also be exposed through an API that accepts relevant borrower and behavioral information and returns a predicted credit-risk probability or classification.

16. Tools and Technologies

The project combines data analytics, machine learning, and business intelligence technologies.

Component	Technology
Programming Language	Python
Data Manipulation	Pandas
Baseline ML Algorithm	Random Forest Classifier
Enhanced ML Algorithm	XGBoost Classifier
Machine Learning Libraries	Scikit-learn, XGBoost
Data Visualization	Microsoft Power BI
Data Storage	CSV, Excel
Development Environment	Visual Studio Code
Model Storage	PKL
Version Control	Git and GitHub

Python and Pandas were primarily used for data preprocessing and behavioral feature engineering. Random Forest and XGBoost were used for credit-risk modelling, while Power BI was used to develop the final analytical dashboard and communicate the model results.

17. Conclusion

This project developed a proof-of-concept framework for alternative-data-enhanced credit-risk assessment using UPI behavioral analytics and machine learning.

The project combined conventional borrower-level credit information with behavioral indicators derived from a dataset containing 250,000 UPI transactions. Transaction-level information was processed and transformed into approximately 400 behavioral segments, incorporating features such as transaction frequency, average transaction amount, transaction volatility, fraud exposure, weekend activity, merchant diversity, and device diversity.

A Random Forest Classifier was implemented as the baseline credit-risk model and achieved a ROC-AUC of 0.7183. The enhanced framework incorporated behavioral information and used an XGBoost Classifier, achieving a ROC-AUC of 0.7457. This represents an absolute ROC-AUC gain of 0.0274, or approximately 3.81% relative improvement over the baseline.

On the reported 160-sample evaluation set, the enhanced XGBoost model correctly classified 155 of 160 samples, resulting in an overall accuracy of 96.88%. For the Poor/risk class, the model achieved 94.03% precision, 98.44% recall, and a 96.18% F1-score. The model also achieved 95.83% specificity. Of the 64 actual Poor cases, 63 were correctly identified, with only one Poor case incorrectly classified as Good.

Feature-importance analysis demonstrated that traditional credit indicators, particularly external credit variables such as EXT_SOURCE_2 and EXT_SOURCE_3, remained among the strongest predictors. At the same time, behavioral variables such as transaction volatility, average transaction amount, fraud ratio, transaction frequency, weekend activity, and merchant diversity provided supplementary information within the enhanced framework.

An interactive three-page Power BI dashboard was developed to communicate the analytical results. The dashboard presents model performance, major credit-risk drivers, UPI behavioral feature importance, key behavioral indicators, and transaction distributions in an interpretable format.

Overall, the project demonstrates how data analytics, alternative digital-payment information, machine learning, and business intelligence can be combined to investigate new approaches to credit-risk assessment.

However, the results must be interpreted within the limitations of the experimental setup. The credit-risk and UPI datasets are not directly linked at the real-customer level, and the baseline and enhanced frameworks use different machine-learning algorithms. Therefore, the observed performance improvement cannot be attributed exclusively to behavioral features.

Future work using correctly linked real-world customer data, controlled model comparisons, explainable AI, temporal behavioral modelling, fairness analysis, model-drift monitoring, and real-time risk assessment could extend the proposed framework towards a more robust financial-risk decision-support system.

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